

Descriptive Analysis of M&A Transactions

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Introduction

Dataset Construction, Objective, & Analysis

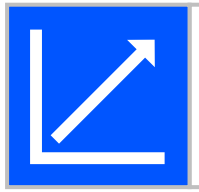
Dataset Construction

We used Python to scrape and build a dataset containing **46 M&A transactions** from publicly available financial data sources. For each transaction, we collected deal characteristics and calculated the acquiring firm's **365-day stock return following the acquisition announcement.**

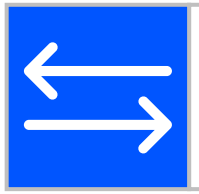
Research Objective

The goal of this analysis is to identify **deal attributes associated with strong post-acquisition shareholder performance**. Specifically, we examine whether certain acquisition characteristics are linked to higher or lower stock returns one year after the transaction.

Key Variables Analyzed



Transformational vs.
Non-Transformational Deals



Same Industry vs.
Cross-Industry Acquisitions



Relative Deal Size
(Small, Medium, Large)

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Dataset Construction

Process Used To Build The Transaction Dataset

1. Deal Identification

A **large language model and manually curated deal list** were used to identify approximately **50 recent acquisitions** for inclusion in the dataset.

2. Python Data Collection

A Python script used the **Yahoo Finance API and Yahoo Finance Search API** to gather financial data for acquiring firms.

3. Return Calculation

For each transaction, the acquiring firm's **365-day stock return following the acquisition announcement** was calculated.

4. Data Preparation

The dataset was cleaned and structured in **Excel**, resulting in **46 usable transactions** where deal characteristics and variables were prepared for the regression analysis.

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

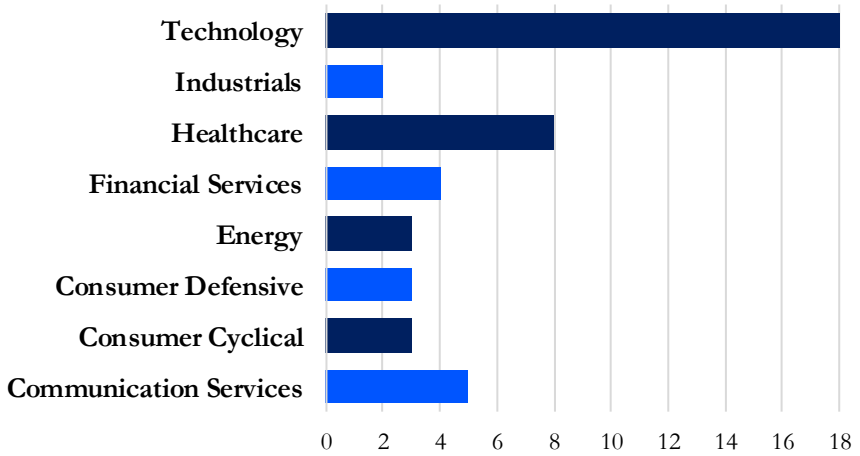
Deal Characteristics

Overview of Sectors, Strategy, & Size

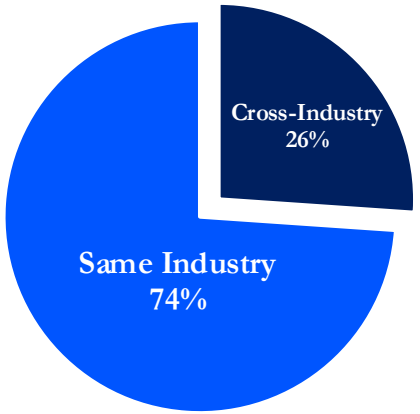
Overview & Characteristics

The dataset includes acquisitions across several industry sectors, with **technology firms** representing the largest portion. Most transactions occur within the **same industry**, and deal sizes are predominantly **small relative to the acquirer**.

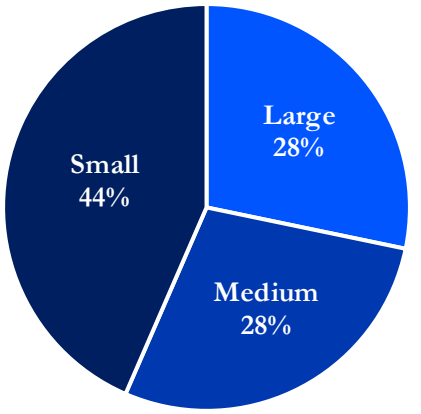
Deal Distr. by Sector



Deal Distr. by Industry Strategy



Deal Distr. by Relative Deal Size



Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Backwards Regression Model

Drivers of Post-Acquisition Returns

The Model

We use a **multiple regression model** to evaluate how **deal characteristics** impact **365-day shareholder returns.**

Equation

$$Return_{365} = \beta_1(Transformative) + \beta_2(Same\ Industry) + \beta_3(Deal\ Size\ Ratio) + \varepsilon(Error\ Term)$$

$Return_{365}$	β_1(Transformative)	β_2(Same Industry)	β_3(Deal Size Ratio)	ε(Error Term)
Percentage change in the acquiring firm's **share price one year following the acquisition.**	Binary indicator for large acquisitions where **deal value exceeds 25%** of the acquirer's **market capitalization.**	Binary indicator for acquisitions where the **acquiring firm and target** operate within the **same industry sector**.	Continuous variable measuring the **ratio of deal value to the acquiring firm's market cap.**	Captures **unobserved factors influencing post-acquisition returns** that are not included in the model.

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Backwards Regression Model

Drivers of Post-Acquisition Returns

First Regression Model

$$Return_{365} = \beta_1(Transformative) + \beta_2(Same\ Industry) + \beta_3(Deal\ Size\ Ratio) + \varepsilon(Error\ Term)$$

Model Performance

$R^2 \approx 0.32$	Model explains about **32% of variation in returns**
Sig. F ≈ 0.001	Overall regression is **statistically significant**

Key Findings

Transformational Deals Reduce Returns ✓
38.5% lower 365-day returns

Industry Alignment Not Significant ✗
Not statistically significant.

Relative Deal Size Not Significant ✗
Not statistically significant

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Backwards Regression Model

Drivers of Post-Acquisition Returns

First Regression Model

$$Return_{365} = \beta_1(Transformative) + \beta_2(Same\ Industry) + \beta_3(Deal\ Size\ Ratio) + \varepsilon(Error\ Term)$$

Model Performance

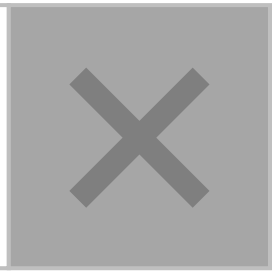
$R^2 \approx 0.30$	Model explains about **30% of variation in returns**
Sig. F ≈ 0.0005	Overall regression is **statistically significant**

Key Findings

Transformational Deals Reduce Returns
41.4% lower 365-day returns



Industry Alignment Not Significant
Not statistically significant.



Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Backwards Regression Model

Drivers of Post-Acquisition Returns

First Regression Model

$$Return_{365} = \beta_1(Transformative) + \beta_2(Same\ Industry) + \beta_3(Deal\ Size\ Ratio) + \varepsilon(Error\ Term)$$

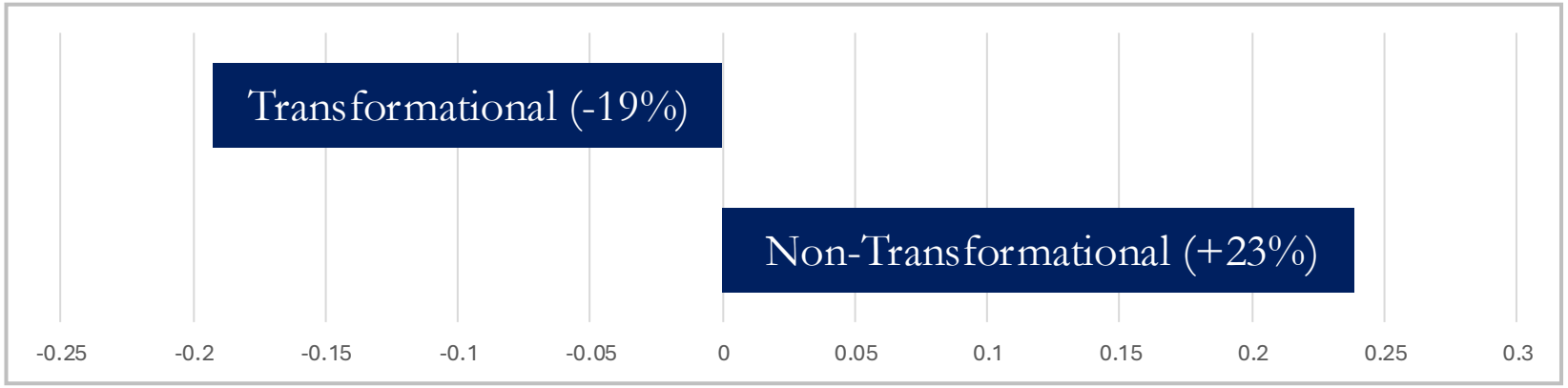
Model Performance

$R^2 \approx 0.27$	Model explains about **27% of variation in returns**
Sig. F ≈ 0.0002	Overall regression is **statistically significant**

Key Findings

Transformational Deals Reduce Returns

43.1% lower 365-day returns

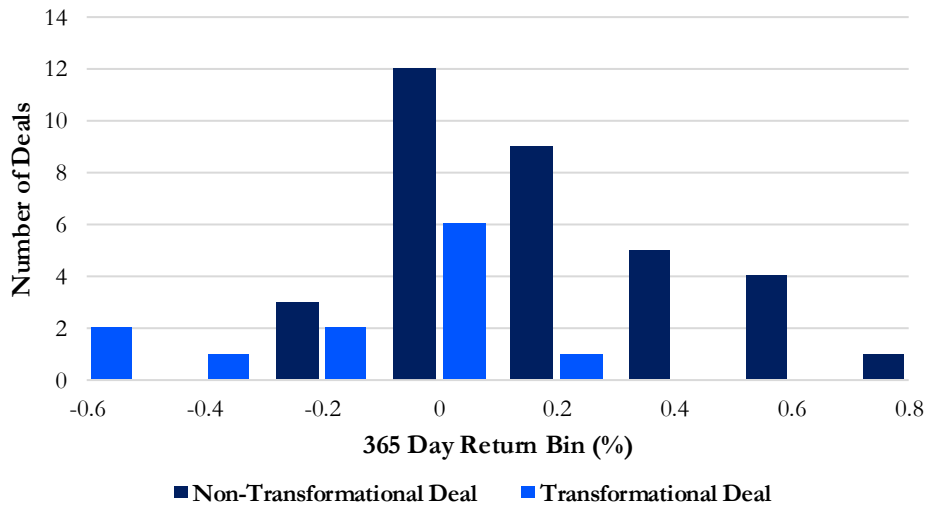


Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

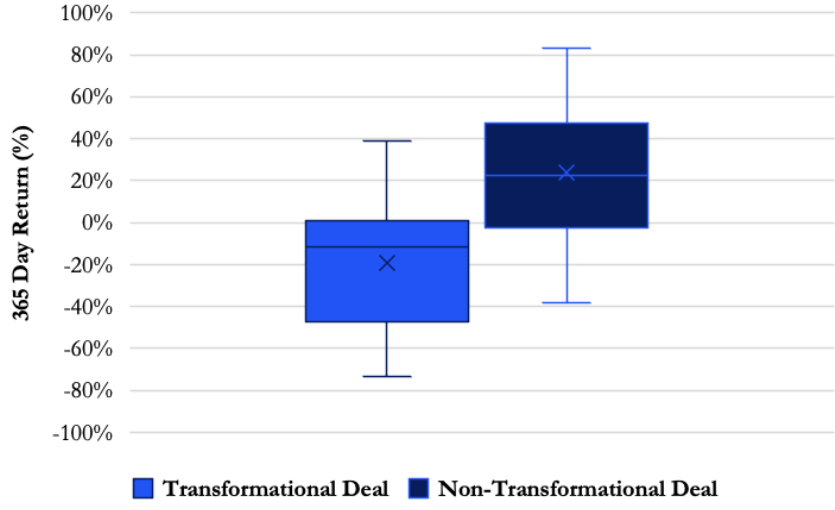
Post Acquisition Returns

Returns & Distr. of Transformational vs. Non-Transformational

Distr. of 365 Day Returns by Deal Type



Post-Acquisition Performance by Deal Type



Return

-43%

Transformational deals show **~43% lower 365-day returns.**

Bottom Line

Transformational acquisitions exhibit **lower median returns** and **greater downside risk** compared to non-transformational deals.

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

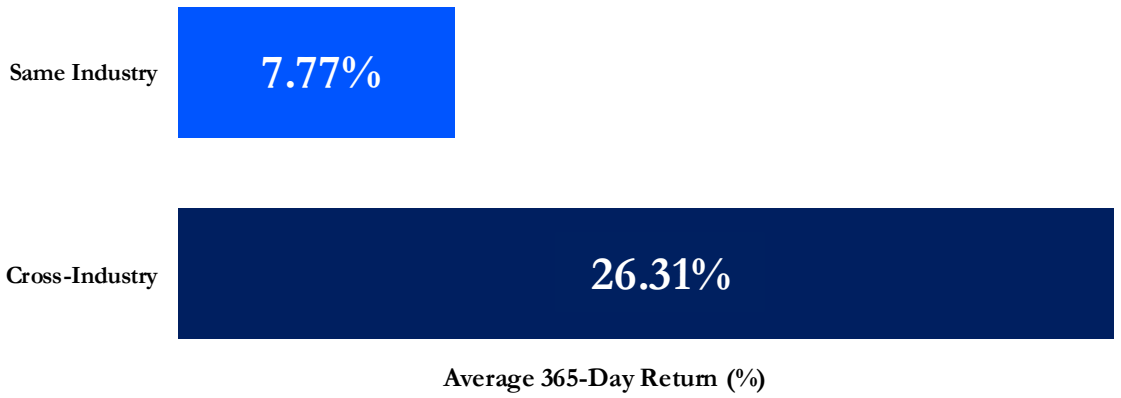
Post Acquisition Returns

Industry Strategy & Relative Deal Size

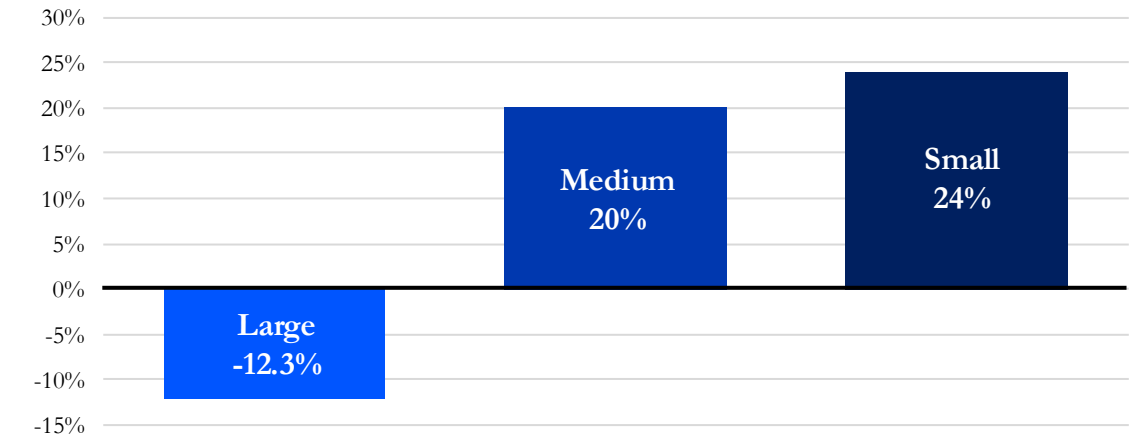
Industry Strategy & Deal Size

Cross-industry acquisitions and **smaller relative deal sizes** exhibit **higher average returns** in the sample; however, these differences **are not statistically significant** in the regression model.

Average Return: Same Industry vs. Cross-Industry



Average Return: Relative Deal Size



Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Limitations of Analysis

Context & Constraints of the Analysis

Limitations

While the model identifies meaningful patterns, acquisition outcomes are influenced by many factors beyond the variables included in this analysis.

Dataset Size



The dataset includes **46 acquisitions**, which limits statistical power and may reduce the generalizability of the results.

Market Conditions



Post-acquisition returns may be influenced by **broader market trends**, not solely the acquisition itself.

Time Horizon



Returns are measured over **365 days**, which may not fully capture the long-term value creation of acquisitions.

Limited Variables



Detailed metrics such as **acquisition premiums or financing structure** were not broadly accessible through free APIs.

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Future Research

Opportunities to Expand & Strengthen the Analysis

1.

Dataset Expansion

Expanding the dataset to **several hundred acquisitions** would improve statistical power and allow for more robust conclusions.

2.

Additional Deal Variables

Future models could incorporate factors such as **acquisition premiums, financing structure, and integration costs** if detailed transaction data becomes available.

3.

Market Controls

Including controls for **overall market performance or sector trends** would help isolate the effect of the acquisition itself on shareholder returns.

4.

Longer Time Horizons

Analyzing returns over **3–5 year periods** may better capture the long-term value creation of acquisitions.

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API

Conclusion

Key Insights From The Analysis

Transformational Acquisitions

Transformational deals are associated with **~43% lower 365-day returns.** This relationship is **statistically significant.**

Industry Alignment

Cross-industry deals show **slightly higher average returns.** However, Regression results indicate **no significant relationship.**

Relative Deal Size

Smaller deals show **somewhat stronger average returns.** However, Regression results indicate **no significant relationship.**

Acquisition Outcomes Are Highly Variable



Post-acquisition returns exhibit **wide dispersion.**

Integration execution and **external factors** likely play a large role.



So What?

The results suggest that **large transformational acquisitions present the greatest risk to post-acquisition shareholder performance**.

Sources: Manually Curated M&A Deal List, Yahoo Finance, Yahoo Finance Search API



Questions?